

DEFENSE STANDARDIZATION PROGRAM

CASE STUDY

Transforming Naval Nuclear Propulsion:
The Impact of the Generic Instrumentation and Control (I&C) Program

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T16:54Z
Check the source to verify that this is the current version before use.

Defense Standardization Program Case Study

Transforming Naval Nuclear Propulsion: The Impact of the Generic Instrumentation and Control (I&C) Program

BACKGROUND

The Naval Nuclear Propulsion Program (NNPP) depends on instrumentation and control (I&C) systems to monitor, control, and protect reactor plants aboard submarines, cruisers, and aircraft carriers. The Navy began transitioning from analog to digital (software-based) I&C systems in the 1980s to improve reactor core utilization, which increased ship speed and core life, simplified operation, and reduced maintenance. By the late 1990s, all operational nuclear warship classes were deploying digital I&C systems but each class developed and qualified unique subsystems despite the high degree of commonality between subsystem functions on all classes of nuclear-powered warships.

Recurring obsolescence management costs grew exponentially due to the increasing number of I&C subsystems and components while non-recurring development and qualification costs of each I&C system suite were approaching $100 million per class.



Figure 1. Generic I&C Systems Use Hundreds Fewer Circuit Cards

THE CHALLENGES

1. Lack of Standardization: I&C systems for each ship class were competitively contracted at different times, tailored to each platform, and rarely reused between classes, increasing costs and complexity.
2. Obsolescence: Digital I&C components have significantly shorter life cycles than analog, resulting in frequent redesign and requalification of circuit cards and software updates. With the growing number of unique components, these activities significantly increased program costs and strained engineering resources.

STANDARDIZATION IN ACTION: HOW THE GENERIC I&C PROGRAM HAS SAVED BILLIONS OF DOLLARS

In response, the NNPP initiated the Generic I&C Program in the late 1990s to reap the benefits of standardization. This program created a universal suite of hardware and software components, including infrastructure, power, and circuit card Hardware Building Blocks (HWBBs).

This program has continued to deliver substantial cost and performance benefits:

- **Non-recurring costs** to develop a reactor plant I&C suite for each warship class were nearly $100 million before the generic I&C suite. Only $40 million (using then-year dollars) was needed for the entire fleet by the time the second generic I&C suite launched, resulting in substantial cost avoidance for the design of new warship classes.
- **Development time** from design through initial fielding for the generic I&C suite was 20% faster than the previous suite: 8 years instead of 10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T16:54Z
Check the source to verify that this is the current version before use.



- **Recurring costs** for materials and services each year were significantly reduced from $10 million per non-generic system to less than $10 million to support the entire fleet.
- **Obsolescence management** became dramatically more efficient. Ships with generic I&C suites do not require technology refreshes during their service lives—a life-cycle savings of $500 million for the VIRGINIA class alone.
- **Operational consistency** improved fleet-wide with standardized training and support processes.
- **Software delivery** became more agile because of centralized infrastructure and the repeated use of proven development and qualification tools.

Non-recurring costs substantially reduced from approximately $100 million for each new warship class to **$40 million for the fleet**

Generic I&C suite designed and **fielded 20% faster, in 8 years instead of 10**

Recurring annual costs are **less than $10 million** to support all nuclear-powered warships, down from **more than $10 million per system**

Streamlined **obsolescence management** with **life-cycle savings of $500 million** for just the VIRGINIA Class Submarines

BROADER IMPACT AND FUTURE OUTLOOK

The Generic I&C Program established a new model for standardization and efficiency in naval systems development, including the following key features:

- A comprehensive suite of specifications and process documents covering everything from software testing to certification planning. These rigorous standards ensure quality and compliance across all stages of development and deployment.
- A significant reduction in total unique circuit cards (from hundreds down to fewer than 25), enabling efficient obsolescence management.
- A shared software development tool chain, operating system, and qualification process that enable centralized software engineering. This approach fostered the creation of a software factory staffed with experienced developers, testers, and engineers.
- A consistent operator interface across the fleet, simplifying training, operation, and maintenance.



Figure 2. Virginia Class Submarine

The success of the Generic I&C Program has drawn international attention. For example, UK naval leaders adopted many of its standards and processes for development, qualification, and obsolescence management.

Next-generation generic I&C suites planned for deployment in the early 2030s are poised to expand beyond traditional applications to support power electronics and large-scale motor drives. The Generic I&C Program demonstrates how disciplined standardization and centralized engineering can transform the affordability, reliability, and scalability of complex military systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T16:54Z
Check the source to verify that this is the current version before use.



All rights reserved. 112025

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T16:54Z
Check the source to verify that this is the current version before use.